

Matching on the Labor and Product Markets

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Product market.

- Households cany v visits
- Firms provide k services (capacity)
- Matching function determines # trades.

$$y = m(k, v)$$

Use CES matching function.

$$y = [k^{-r} + v^{-r}]^{-\frac{1}{r}} \quad r > 0$$

- Each visit cost $p > 0$ resources

Labour market

- Firms post $\hat{v}$ vacancies
- There are h workers in labour force $\rightarrow$ all initially unemployed.
- Matching function determines # of hires.

$$l = \hat{m}(h, \hat{v})$$

Use CES matching function:

$$l = [h^{-\hat{r}} + \hat{v}^{-\hat{r}}]^{-\frac{1}{\hat{r}}} \quad \hat{r} > 0$$

- Each vacancy requires $\hat{p}$ resources

Market tightness:

- Product market tightness.

$$\alpha = v / k$$

- Labour market tightness:

$$\theta = \hat{v} / h$$

Trading probabilities

- Buying probability $q(x)$
- Selling probability $f(x)$
- Recruiting probability $\hat{q}(\theta)$
- Job-finding probability $\hat{f}(\theta)$

Expression of probabilities

$$\hat{f}(\theta) = \frac{c}{h} = \frac{[h^{-\hat{r}} + \bar{v}^{-\hat{r}}]^{-1/\hat{r}}}{h} = [1 + \theta^{-\hat{r}}]^{-1/\hat{r}}$$

$$\hat{q}(\theta) = \frac{c}{\bar{v}} = \frac{[h^{-\hat{r}} + \bar{v}^{-\hat{r}}]^{-1/\hat{r}}}{\bar{v}} = [\theta^{\hat{r}} + 1]^{-1/\hat{r}}$$